



Syllabus Application



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CS 306 Database Systems

Faculty Faculty of Engineering and Natural Sciences

Semester Fall 2025-2026

Course CS 306 - Database Systems

Time/Place	<u>Time</u>	<u>Week Day</u>	<u>Place</u>	<u>Date</u>
	16:40-17:30	Thu	SBS-G071	Sep 29, 2025-Jan 3, 2026
	12:40-14:30	Fri	FENS-G077	Sep 29, 2025-Jan 3, 2026

Level of course Undergraduate

Course Credits SU Credit:3, ECTS:6, Engineering:6

Prerequisites CS 204 or DSA 201

Corequisites CS 306R

Course Type Lecture

Instructor(s) Information

Yücel Saygın

Email: ysaygin@sabanciuniv.edu

Course Information

Catalog Course Description

This course covers the fundamental topics in database management systems including ANSI/SPARC architecture, data abstraction, data independence; data definition and manipulation languages; entity-relationship and relational data models, integrity constraints, query languages like SQL, relational algebra and calculus, functional dependencies, normal forms, transaction processing recovery and concurrency control, data structure for database management systems such as indices, B-trees, hashing.

Learning Outcomes:

1.	After the course, the student will be able to develop a database application starting with the conceptual design leading to logical and physical designs.
2.	After the course, the student will be competent on relational model and SQL.
3.	After the course, the student will be able to implement a database application based on a given design.
4.	After the course, the student will be able explain the concept of functional dependency and will be able to avoid redundancies and other problems causing by bad designs.
5.	After the course, the student will be able to explain how a DBMS concurrently handles the transactions and what happens after a system crash.
6.	After the course, the student will be able to explain how the data is physically stored, managed, and queried within the database.

Course Objective

To supply the students with the theory and practice of database management systems, and database application development.

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Course Materials

Resources:

Technology Requirements:

Computers

Policies

SU Academic Integrity Statement:

<https://www.sabanciuniv.edu/en/academic-integrity-statement>